

Table 1: Case 1 boundary velocity convergence (1 of 2).

h	$l = 2, m = 1$				$l = 2, m = 2$				$l = 4, m = 2$			
	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\mathbf{u})$	rate
1/8	8.877×10^{-2}	–	8.511×10^{-2}	–	8.782×10^{-2}	–	8.478×10^{-2}	–	8.760×10^{-2}	–	8.756×10^{-2}	–
1/16	5.788×10^{-3}	3.94	5.496×10^{-3}	3.95	5.786×10^{-3}	3.92	5.487×10^{-3}	3.95	5.942×10^{-3}	3.88	6.265×10^{-3}	3.80
1/32	2.762×10^{-4}	4.39	3.142×10^{-4}	4.13	2.764×10^{-4}	4.39	3.145×10^{-4}	4.12	3.383×10^{-4}	4.13	5.312×10^{-4}	3.56
1/64	1.653×10^{-5}	4.06	5.257×10^{-5}	2.58	1.654×10^{-5}	4.06	5.257×10^{-5}	2.58	2.984×10^{-5}	3.50	9.674×10^{-5}	2.46

Table 2: Case 1 boundary velocity convergence (2 of 2).

h	$l = 4, m = 4$				$l = 8, m = 4$				$l = 8, m = 8$			
	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\mathbf{u})$	rate
1/8	8.839×10^{-2}	–	8.836×10^{-2}	–	8.885×10^{-2}	–	1.018×10^{-1}	–	8.986×10^{-2}	–	1.049×10^{-1}	–
1/16	5.913×10^{-3}	3.90	6.234×10^{-3}	3.83	6.205×10^{-3}	3.84	8.936×10^{-3}	3.51	6.170×10^{-3}	3.86	8.911×10^{-3}	3.56
1/32	3.382×10^{-4}	4.13	5.296×10^{-4}	3.56	4.244×10^{-4}	3.87	1.069×10^{-3}	3.06	4.239×10^{-4}	3.86	1.072×10^{-3}	3.05
1/64	2.976×10^{-5}	3.51	9.631×10^{-5}	2.46	5.715×10^{-5}	2.89	2.117×10^{-4}	2.34	5.713×10^{-5}	2.89	2.100×10^{-4}	2.35

Table 3: Case 1 boundary radial-stress convergence (1 of 2).

h	$l = 2, m = 1$				$l = 2, m = 2$				$l = 4, m = 2$			
	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\sigma_{rr})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\sigma_{rr})$	rate	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\sigma_{rr})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\sigma_{rr})$	rate	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\sigma_{rr})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\sigma_{rr})$	rate
1/8	1.500×10^{-1}	–	9.413×10^{-2}	–	1.458×10^{-1}	–	9.370×10^{-2}	–	1.424×10^{-1}	–	1.073×10^{-1}	–
1/16	1.096×10^{-2}	3.77	8.041×10^{-3}	3.55	1.095×10^{-2}	3.74	8.024×10^{-3}	3.55	1.054×10^{-2}	3.76	1.272×10^{-2}	3.08
1/32	2.905×10^{-4}	5.24	1.100×10^{-3}	2.87	2.920×10^{-4}	5.23	1.102×10^{-3}	2.86	2.737×10^{-4}	5.27	2.392×10^{-3}	2.41
1/64	6.125×10^{-5}	2.25	3.258×10^{-4}	1.76	6.364×10^{-5}	2.20	3.276×10^{-4}	1.75	9.848×10^{-5}	1.47	6.990×10^{-4}	1.77

Table 4: Case 1 boundary radial-stress convergence (2 of 2).

h	$l = 4, m = 4$				$l = 8, m = 4$				$l = 8, m = 8$			
	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\sigma_{rr})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\sigma_{rr})$	rate	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\sigma_{rr})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\sigma_{rr})$	rate	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\sigma_{rr})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\sigma_{rr})$	rate
1/8	1.447×10^{-1}	–	1.083×10^{-1}	–	1.489×10^{-1}	–	1.618×10^{-1}	–	1.499×10^{-1}	–	1.637×10^{-1}	–
1/16	1.052×10^{-2}	3.78	1.270×10^{-2}	3.09	1.203×10^{-2}	3.63	3.009×10^{-2}	2.43	1.192×10^{-2}	3.65	2.971×10^{-2}	2.46
1/32	2.683×10^{-4}	5.29	2.379×10^{-3}	2.42	6.680×10^{-4}	4.17	6.678×10^{-3}	2.17	6.658×10^{-4}	4.16	6.670×10^{-3}	2.16
1/64	9.866×10^{-5}	1.44	6.942×10^{-4}	1.78	1.547×10^{-4}	2.11	1.841×10^{-3}	1.86	1.547×10^{-4}	2.11	1.831×10^{-3}	1.87